

7(a)	photon absorbed (by electron) and electron excited	B1
	photon energy equal to difference in (energy of two) energy levels	B1
	photon energy relates to a single wavelength / single frequency	B1
	electron de-excites and emits photon in any direction	B1
7(b)	$\frac{hc}{\lambda} = \Delta E$	C1
	uses 658nm	C1
	$\frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{658 \times 10^{-9}} = -E_1 - (-3.40 \times 1.60 \times 10^{-19})$ $E_1 = -2.42 \times 10^{-19} \text{ J}$	A1